

Jobs $\{J_a, J_b, J_c, J_d\}$

Precedence constraints

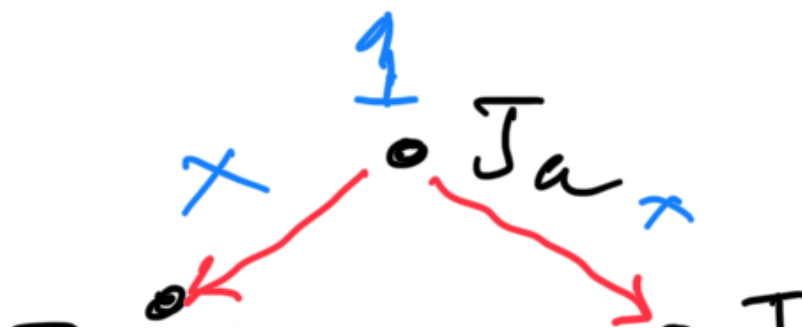
$$J_a < J_b$$

$$J_a < J_d$$

$$J_d < J_c \quad J_c < J_b$$

Define an ordering of jobs to satisfy constraints

$J_a \quad J_d \quad J_c \quad J_b \quad \checkmark$



before → after



$$f: V \rightarrow \{1, 2, \dots, n\}$$

$$J_\alpha \prec J_\beta \Rightarrow f(J_\alpha) < f(J_\beta)$$

When does such an ordering exist

Such an ordering is possible iff there is no cycle

—

In a DAG there is at least one vertex with indegree 0 (source) and at least one vertex with outdegree 0 (sink)

Topological sort

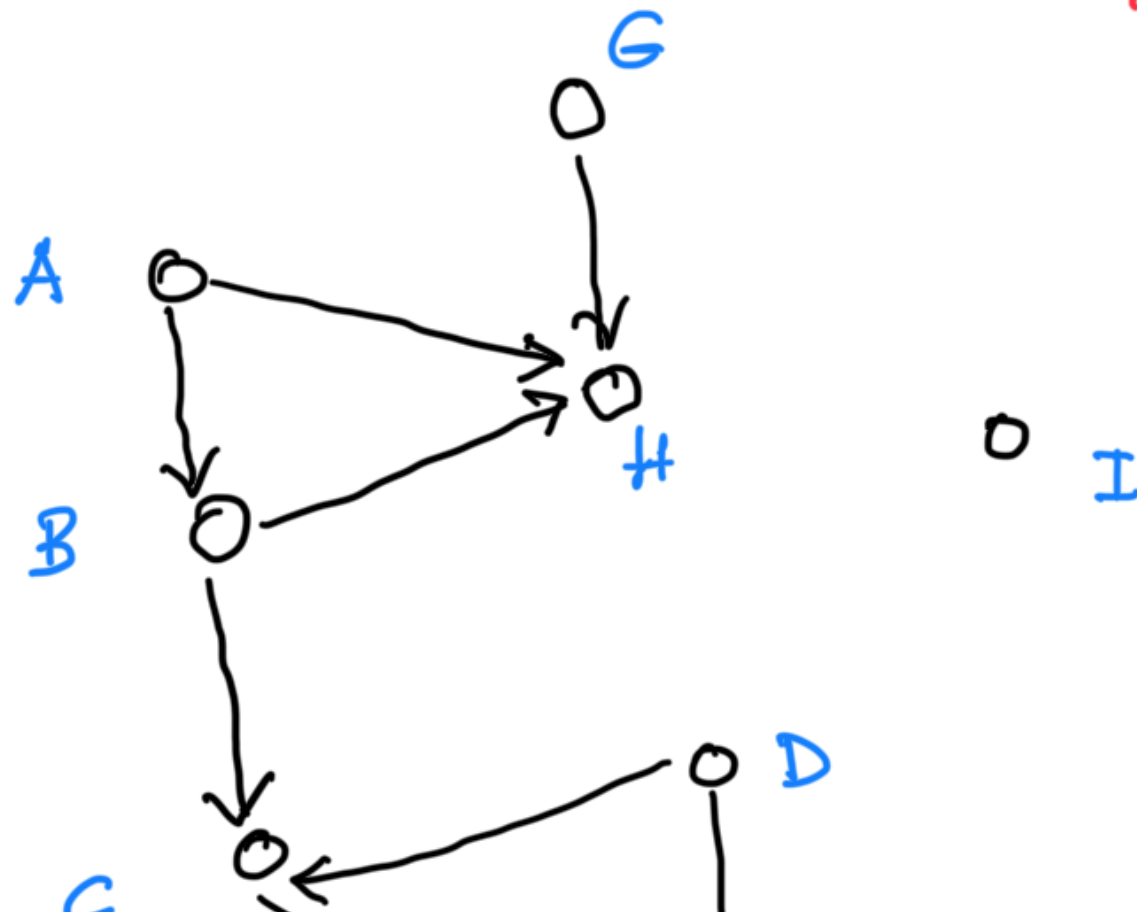
Observation: Among the remaining jobs,

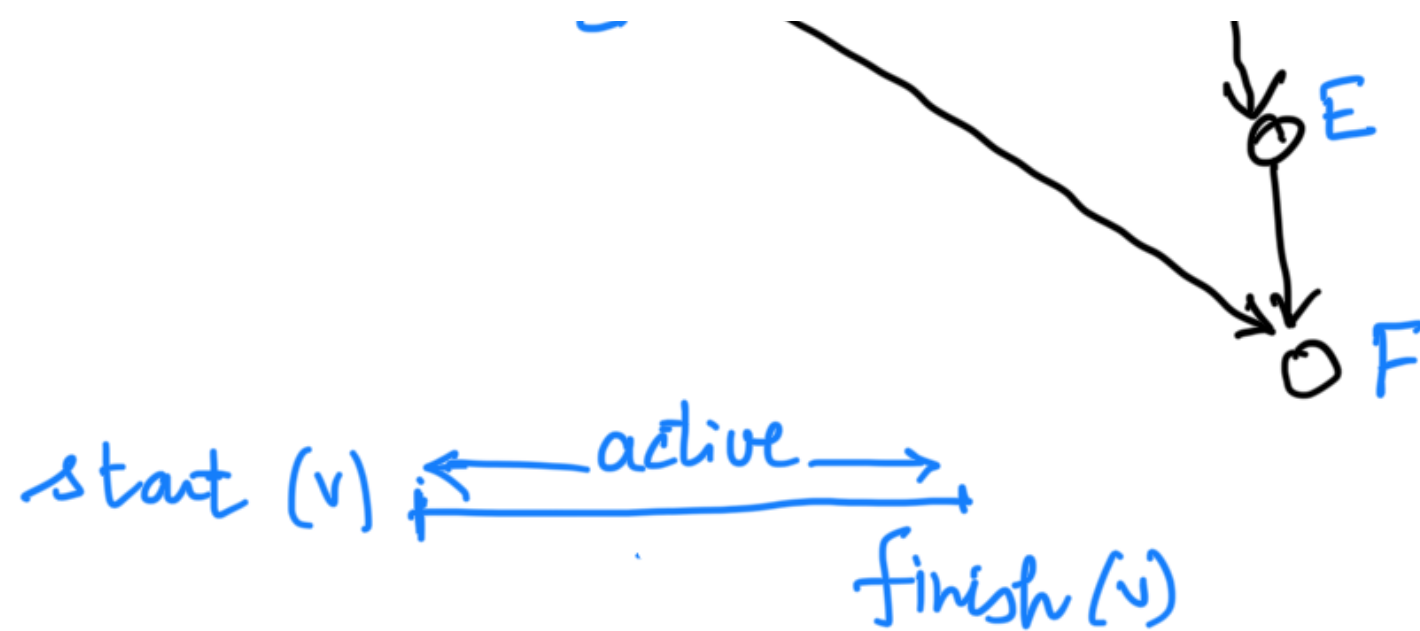
(the source should be scheduled first
(multiple sources - break ties arbitrarily)
(Delete all outgoing edges)

Claim : This numbering produces a feasible schedule

Proof : The source has no prior jobs to
be scheduled, so it is the first job

This observation can be applied to the remaining
graph, since it is also a DAG.
No constraint is violated





$$s(u) \quad s(w) \quad f(u) \quad f(w)$$

Not possible

$$s(u) \quad f(u) \quad s(w) \quad f(w)$$

$\Rightarrow u \rightsquigarrow w$

$$s(u) \quad s(w) \quad f(w) \quad f(u)$$

$\Rightarrow u \rightsquigarrow w$

Claim If $u \rightsquigarrow w \Rightarrow f(u) > f(w)$

Case : $s(u) < s(w) \mid s(u) > s(w)$

Can finishing times provide a soln to the job scheduling problem?

As a corollary of the observation, we can obtain a feasible ordering by reversing the order of the finish times (using radix sort for integers $\in [1, \dots, 2N]$)

Strongly Connected Components

Maximal subsets of vertices $W \subseteq V$ s.t.

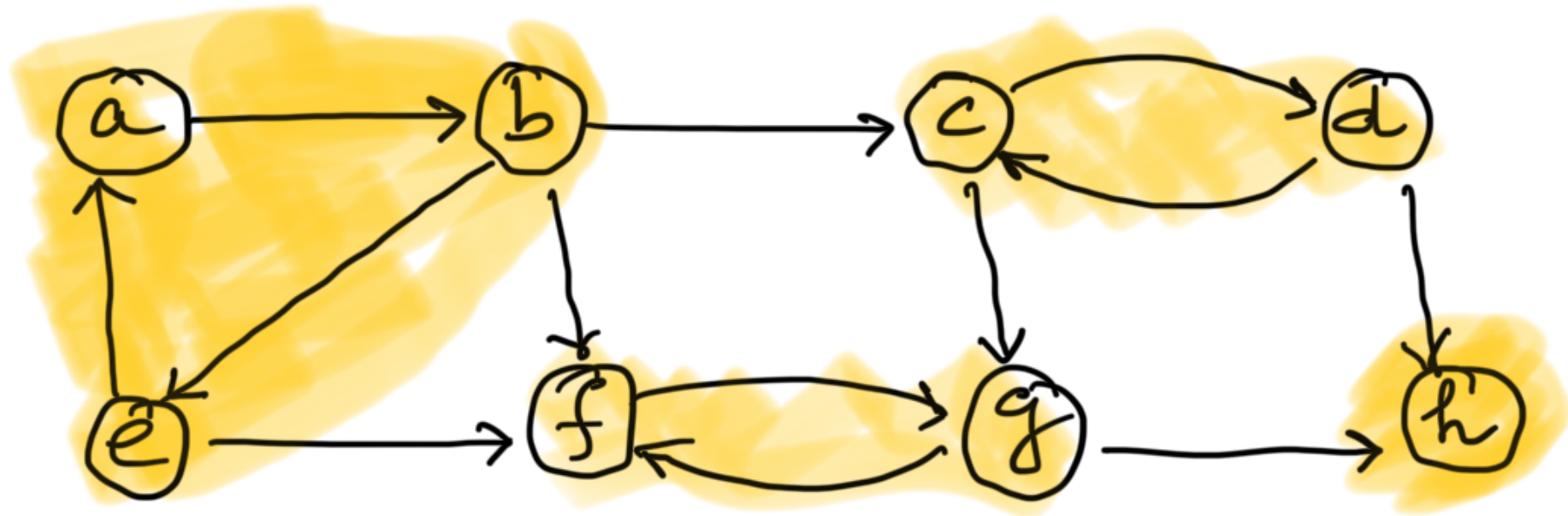
$\forall x, y \in W \quad x \rightsquigarrow y \text{ and } y \rightsquigarrow x$

Claim: SCC defines an equivalence class

based on $(x \approx y)$ if $x \rightsquigarrow y$ and $y \rightsquigarrow x$

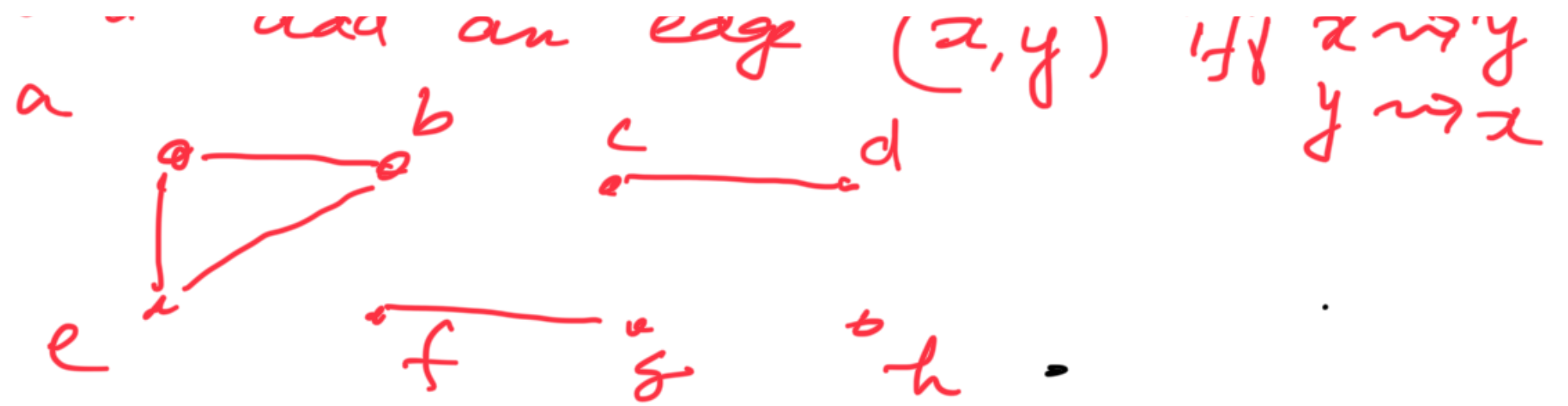
Proof $\approx$ is reflexive, symmetric & transitive

Analogous to connectivity reln in undirected graphs

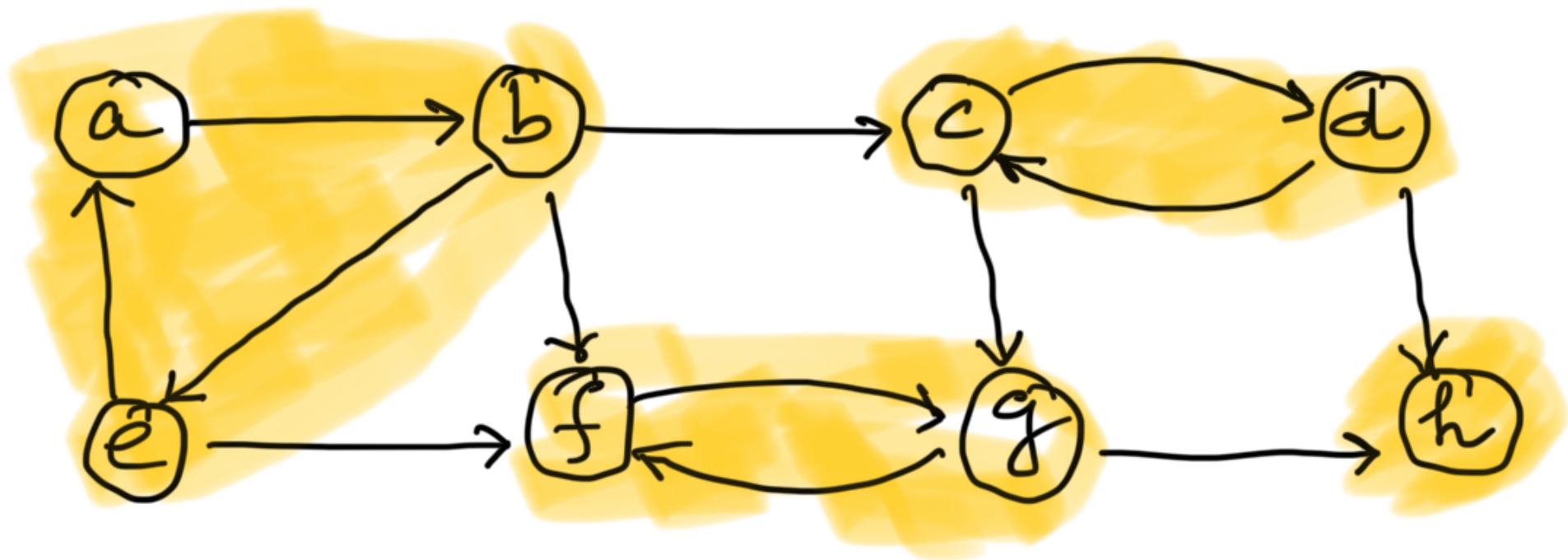


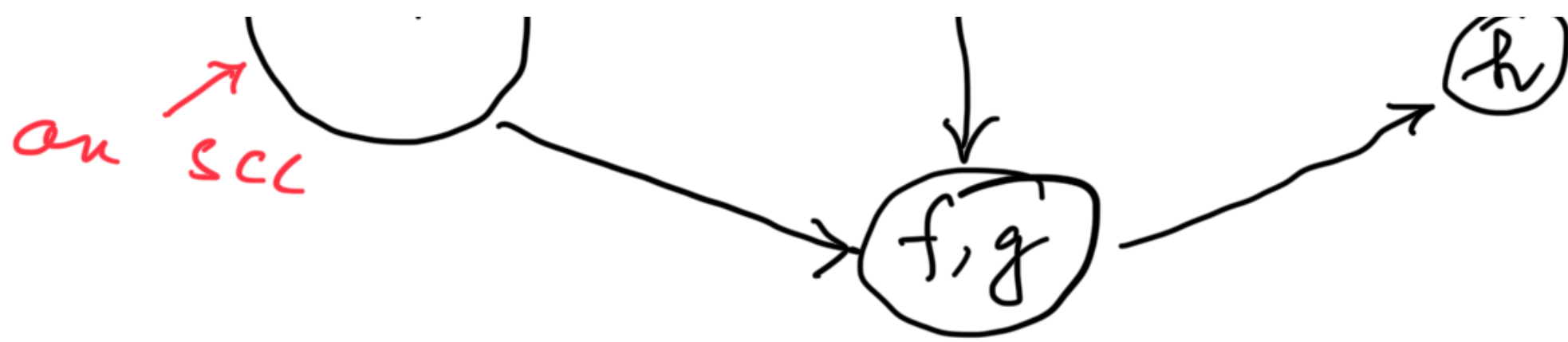
How to find
SCC?

Define an undirected graph
with same set of vertices
and add an undirected edge between



Claim: The connected component corresponds to SCC of the original graph

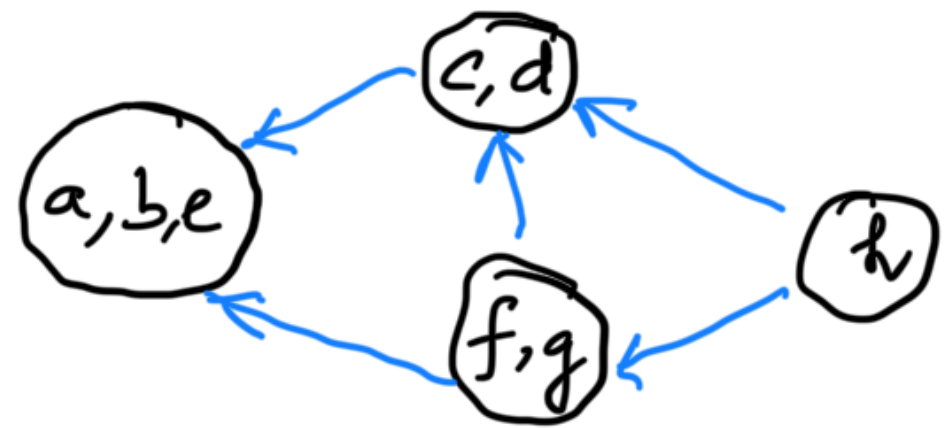
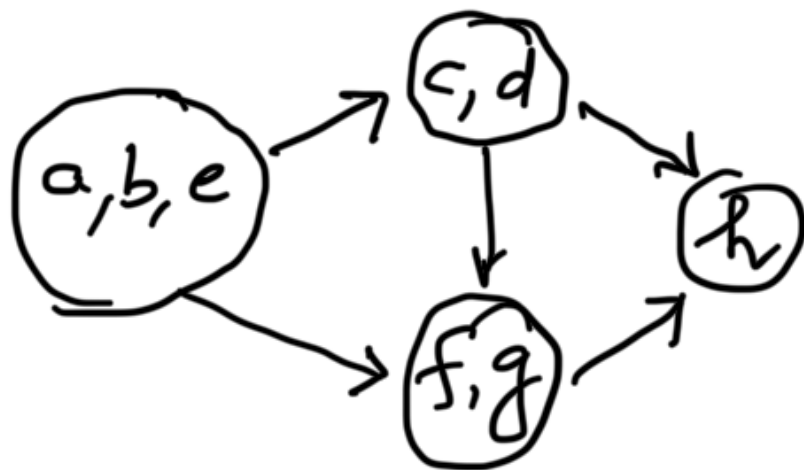
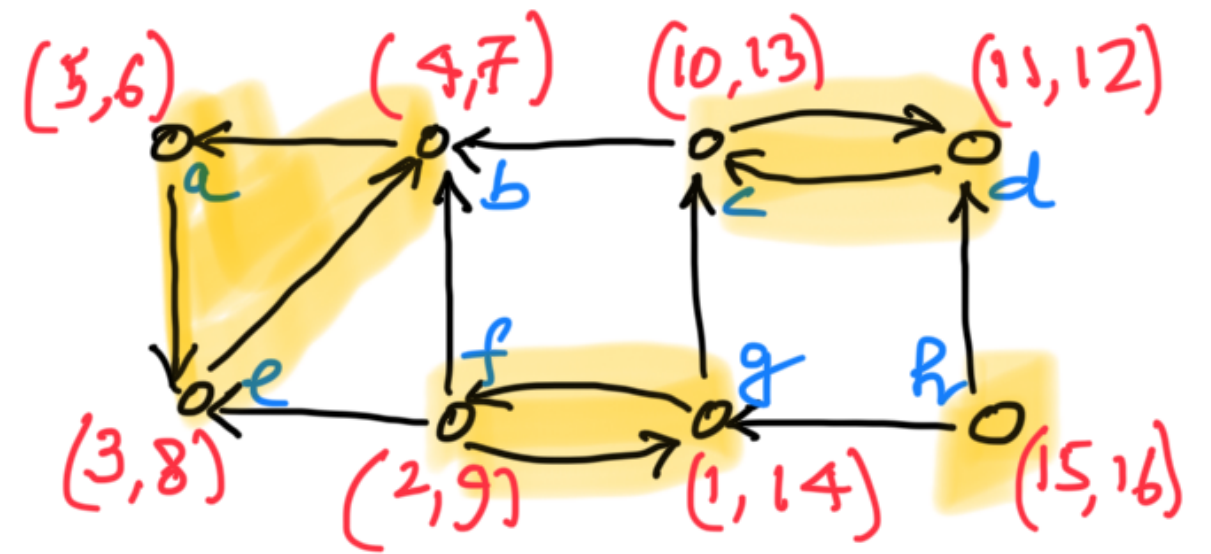
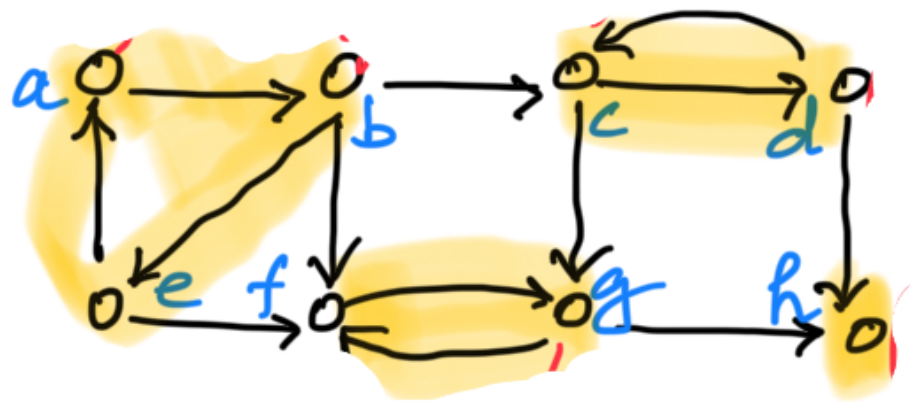




Idea (1) If we do a DFS in a "sink" component, we will reach exactly those vertices (and nothing more)

(2) After deleting all vertices in the sink component, we can repeat the same in the remaining graph (that must have a sink)

— How do we identify a sink component?



Algorithm

Algorithm SCC

DFS (G^R) .

Let $f'(v_i)$ be the finishing times

Initialise $V' = V$

while $V' \neq \emptyset$ do

[$s = \operatorname{argmax}_{x \in V'} f(x)$
Output all vertices W reachable
from s as the next SCC
Delete W